

Mohamed Khider University – Biskra Computer Science Module and Level: CA1-1 st year Engineer Date : 13/01/2025 Time : 01h :30	EXAM	<u>Note:</u>
Calculator and mobile phone are not allowed ممنوع استخدام الآلة الحاسبة والهاتف النقال		
Full Name Group :		

Exercise 1 – (5pts)

1. Make the following conversions :

Decimal	Binary	Octal	Hexadecimal
22,250			
$16^2 + 2^5 + 2^2 + 16^{-1}$			
		75,6	
			2CE
$(11110111)_2 = (\dots\dots\dots)_{\text{Gray}}$		$(010001011100)_{\text{Exces3}} = (\dots\dots\dots)_{10}$	

2. Do the sum and correct the results: $(86)_{10} + (49)_{10}$

<u>In BCD code</u>	<u>In XS3 code</u>

Exercise 2 – (5pts)

1. Give in 9 bits the representations of the following decimal values:

Decimal	SAV	1'sC	2's C
+23			
			111011000

2. Perform the following operation in 2'sC on 7bits and comment result :

$(-55)_8 + (23)_8$	$(+25)_{16} + (2E)_{16}$
+ _____ =	+ _____ =

3. Let the decimal number x = 3.15, Convert the number x to floating point according to the IEEE 754 (32 bits)?

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4. Convert to decimal the number: $(42EA8000)_{16}$, encoded in IEEE754 single-precision format.

Binary number								
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Exercise 3 - (6pts)

Let the function F1 be defined by the following truth table:

x	y	z	F
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

1. Give its disjunctive canonical form (1st CF):

2. Demonstrate the following form : $\overline{((\bar{a} + \bar{b}) \cdot a \cdot b)} = 1$

3. Simplify this function F1 by the algebraic method and draw the logic diagram:



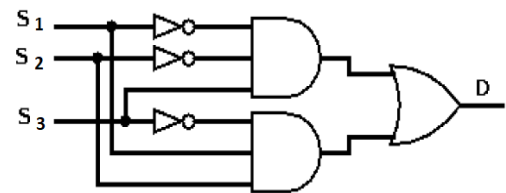
4. Simplify F using the Karnaugh method:

$$F(A,B,C,D) = \sum (0, 1, 3, 5, 7, 9, 10, 11, 13, 15)$$

$$F(A,B,C,D) = \dots\dots\dots$$

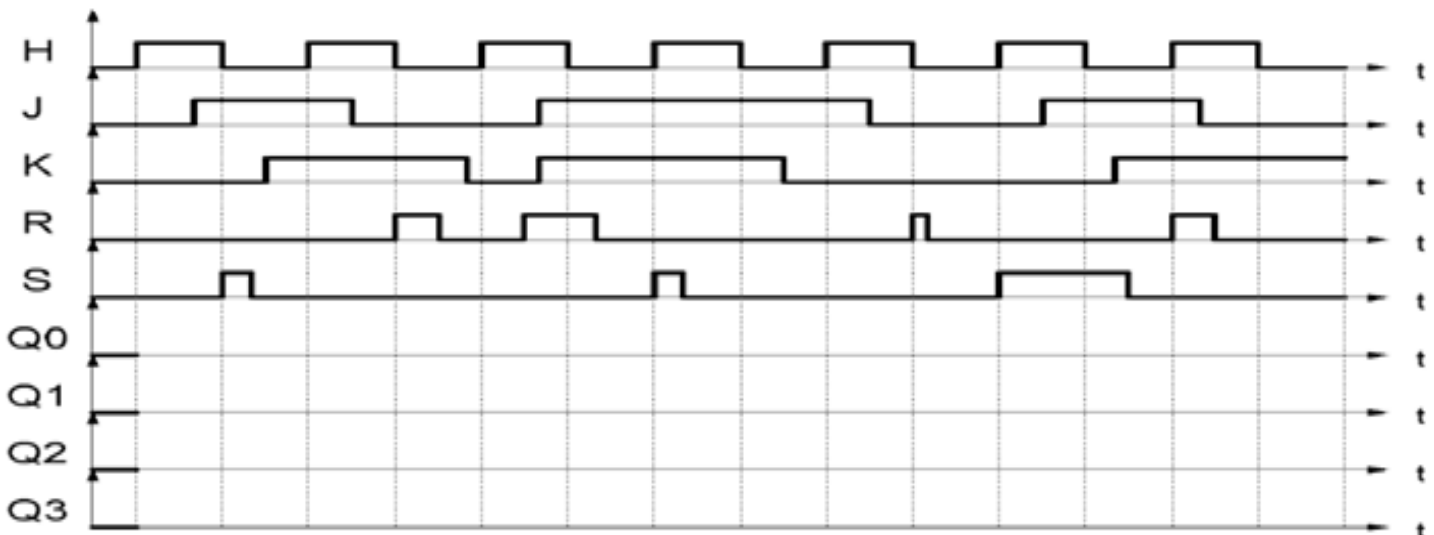
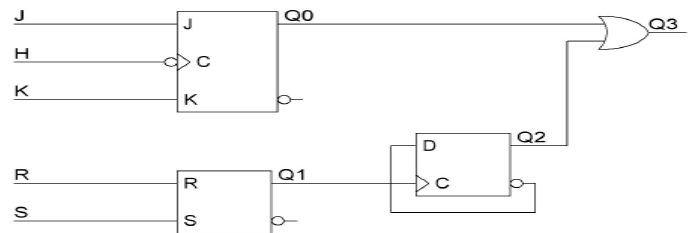
5. Draw the new logic circuit of D:

$$D(S_1, S_2, S_3) = \dots\dots\dots$$



Exercise 4 - (4pts)

Fill the chronogram according to the following montage :



Good Luck